

Tide retrospective: concrete `GibbsRegularity` for the harmonic Gibbs

Daniel Murfet

6 May 2026

Abstract

This retrospective records Tide step 10, the second tide of the day, which provides a concrete `GibbsRegularity` instance for the 1D harmonic potential $L(x) = (\lambda/2)x^2$ with the linear perturbation $A(x) = x$ on $\mathbb{R}$ ($\lambda, t > 0$). The instance unlocks threepoint’s *derivative* theorems — the FDT, cross-susceptibility, and flow equation — for the harmonic case, transforming them from parameterised lemmas (conditional on a hypothesis bundle) into theorems about the harmonic Gibbs measure proper. The formalisation took 265 lines of Lean in a new file `Laplace/OneD/HarmonicGibbsRegularity.lean`, with zero `sorry` / `axiom` / `native_decide`, and landed in a single session. The single decisive contribution from the GPT-5.5 Pro deliberation was a strategic pivot away from the expected dominated-convergence approach (`hasDerivAt_integral_of_dominated_loc.*`) toward a *closed-form* proof of the partition function for all h , which sidesteps the differentiation-under-the-integral machinery entirely and produces a stronger primitive theorem as a free side effect.

1 Setting

The `kappa3`-harmonic tide (Tide 5) and the anharmonic- κ_3 tide (Tide 9) both deliberately avoided `GibbsRegularity`. The reason was given in Tide 5’s GPT consult, quoted verbatim in that tide’s retrospective: “*`kappa3` is defined using `gibbsExp ... 0 ...`, i.e. at $h = 0$. To show $\kappa_3(x, x, x) = 0$ you only need $\langle x \rangle = 0$, $\langle x^3 \rangle = 0$. You do not need the FDT differentiation machinery.*” Both tides therefore proved their cumulant statements without instantiating the regularity bundle.

The cost of that decision was that threepoint’s *derivative* theorems — the FDT ($\partial_h \langle \varphi \rangle_h \upharpoonright_{h=0} = -t \cdot \text{Cov}(\varphi, A)$), the cross-susceptibility identity ($\partial_h \text{Cov}_h(\varphi, B) \upharpoonright_{h=0} = -t \cdot \kappa_3(\varphi, A, B)$), and the flow equation — remained *parameterised* on the hypothesis bundle. They are theorems of the form “if you give me a `GibbsRegularity` instance for (L, A, t) , here’s the FDT” rather than “here’s the FDT for the harmonic Gibbs.”

The 6 May candidates survey listed providing the harmonic instance as **C3**: “the natural follow-up” to Tide 5, with an estimated size of 150–250 lines. The expected approach was to apply Mathlib’s `hasDerivAt_integral_of_dominated_loc_of_lip` (or its derivative-bound variant) with a square-completion dominator on a neighbourhood of $h = 0$.

2 The thing formalised

The headline statement is:

For $\lambda > 0$ and $t > 0$, the triple $(\mu, L, A) = (\text{volume}, (\lambda/2)x^2, x)$ on $\mathbb{R}$ satisfies the regularity bundle:

`Threepoint.GibbsRegularity (volume) ( $\lambda/2 \cdot \square^2$ ) (id) t.`

The Lean signature:¹

```
theorem Threepoint.harmonic_id_gibbsRegularity
  {lam t : R} (hlam : 0 < lam) (ht : 0 < t) :
  Threepoint.GibbsRegularity (volume : Measure R)
    (fun x : R => lam / 2 * x ^ 2)
    (fun x : R => x) t
```

(Writing R for the type of real numbers, which the Lean source spells $\mathbb{R}$.)

The instance discharges three fields: strict positivity of the unperturbed partition, the $h = 0$ identity (a definitional simp-reduction), and differentiability of the partition under the integral at $h = 0$ with the explicit pointwise-derivative integral as the derivative value.

The headline theorem is named in the `Threepoint` namespace (via the `_root_` prefix²) even though it lives in the `laplace` repo. This is consistent with the cross-repo architecture established by Tide 9 and recommended by GPT: `laplace` depends on `threepoint`, and the `harmonic` instance for `threepoint`'s structure is naturally proven where `laplace`'s asymptotic machinery is available, while remaining namespace-discoverable as a `threepoint`-side fact.

3 Proof strategy

The proof structure is shorter than the dominated-convergence approach would have been. The pivot to the closed-form route was the decisive Step 2 contribution.

The closed form. For any $h \in \mathbb{R}$ and $\lambda, t > 0$,

$$Z(h) := \int_{\mathbb{R}} e^{-t((\lambda/2)x^2 + hx)} dx = e^{th^2/(2\lambda)} \sqrt{2\pi/(\lambda t)}.$$

The proof is a four-step calculation:

1. *Square completion*, as a pointwise identity:

$$(\lambda/2)x^2 + hx = (\lambda/2)(x + h/\lambda)^2 - h^2/(2\lambda).$$

2. *Boltzmann factor decomposition*:

$$e^{-t((\lambda/2)x^2 + hx)} = e^{th^2/(2\lambda)} \cdot e^{-(t(\lambda/2))(x+h/\lambda)^2}.$$

3. *Translation invariance* of Lebesgue measure (`integral_add_right_eq_self`, generated by `to_additive` from `integral_mul_right_eq_self`):

$$\int e^{-(t(\lambda/2))(x+h/\lambda)^2} dx = \int e^{-(t(\lambda/2))y^2} dy.$$

4. *Mathlib's integral_gaussian* at $b = t\lambda/2$:

$$\int e^{-by^2} dy = \sqrt{\pi/b} = \sqrt{2\pi/(\lambda t)}.$$

¹File: `Laplace/OneD/HarmonicGibbsRegularity.lean` on `tide/harmonic-gibbsregularity` at commit `3e95014`.

²Lean syntax: `_root_.Threepoint.harmonic_id_gibbsRegularity`.

The three regularity fields, from the closed form.

- *Partition positivity.* $Z(0) = \sqrt{2\pi/(\lambda t)}$, and $\sqrt{\cdot}$ of a positive real is positive (`Real.sqrt_pos`).
- *The $h = 0$ identity.* $\int e^{-t((\lambda/2)x^2 + 0 \cdot x)} dx = \int e^{-t(\lambda/2)x^2} dx$ by integrand congruence (the $0 * x$ simplifies to 0, and $(\dots) + 0 = (\dots)$).
- *Differentiability at $h = 0$.* The partition function in h equals the closed-form expression $h \mapsto e^{th^2/(2\lambda)} \cdot \sqrt{2\pi/(\lambda t)}$. The derivative of this expression at $h = 0$ is zero, since $\frac{d}{dh}[th^2/(2\lambda)] = th/\lambda$ vanishes at $h = 0$, and $\exp(0) \cdot 0 = 0$. The target derivative value $\int (-t \cdot x) e^{-(t\lambda/2)x^2} dx$ is also zero by parity. So both sides equal zero, and `HasDerivAt` closes by the closed-form differentiation.

The mathematical content is short. Most of the 265 lines are Lean plumbing: rewriting integrand lambdas, congruence lemmas, the `HasDerivAt.comp` chain for the closed-form derivative.

4 Roadblocks and resolutions

The strategic pivot. The pre-deliberation plan was to apply Mathlib’s dominated-convergence theorem for differentiation under the integral, with a square-completion dominator.³ GPT named this *Candidate A* and replaced it with *Candidate A'*: the closed-form route. The verbatim recommendation:

GPT-5.5 Pro: Short version: I would **not** lead with dominated differentiation for this instance. For the concrete harmonic+linear case, the cleanest proof is usually: (1) prove the closed form $Z(h) = e^{th^2/(2\lambda)} \sqrt{2\pi/(\lambda t)}$ by completing the square + translation invariance of `volume`; (2) deduce `partition_pos` and `partition_h_zero`; (3) get `partition_hasDerivAt` from the derivative of the closed form, and show the target integral is 0 by oddness. ... That is both stronger and likely shorter than the dominated proof.

The key insight is that for the harmonic Gibbs the partition function is computable in closed form, so there is no need to invoke the abstract differentiation theorem at all. The closed form is also a stronger primitive — it gives the value of $Z(h)$ for all h , not just at $h = 0$ — and is genuinely useful for downstream work (e.g. moment-generating function calculations) in a way the bare `GibbsRegularity` instance is not.

The arithmetic correction. GPT also caught a minor arithmetic error in the pre-deliberation Young bound: Claude wrote $(\lambda/2)x^2 + hx \geq (\lambda/4)x^2 - h^2/(2\lambda)$ when the correct bound is $\geq (\lambda/4)x^2 - h^2/\lambda$ (factor of two off in the second term). This bound becomes moot under the closed-form route, but is worth recording for any future tide that does *do* dominated convergence on a non-Gaussian potential where the Young inequality doesn’t dissolve into a closed form.

The `conv_lhs => ext x; rw [...]` non-tactic. A first attempt at rewriting the integrand inside an integral via `conv_lhs` failed: `ext x` doesn’t dig into integral expressions, only into function-typed targets. The error was “invalid ‘ext’ conv tactic, function or arrow expected: $\int (x : \mathbb{R}), \dots : \mathbb{R}$ ”. The fix was to prove the integrand-equality at the lambda level ($f = g$ as functions) and then `rw` the resulting integral equality. Roughly:

³The two relevant Mathlib lemmas live in `ParametricIntegral.lean`: one takes a Lipschitz bound, the other takes a derivative bound. Either suffices for the harmonic case.

```

have h_integrand_eq : (fun x => ...) = (fun x => ...) := by
  funext x; <ring/exp_add manipulation>
rw [show (integral x, ...) = (integral x, ...)
    from by rw [h_integrand_eq]]

```

This is a recurring Lean idiom for “rewrite the integrand of an integral” and is worth landing in the laplace CLAUDE.md.⁴

hasDerivAt_pow type ascription. Mathlib’s `hasDerivAt_pow` is generic over the field: it takes a natural-number exponent and a base in any normed field $\mathbb{K}$. Calling it as `hasDerivAt_pow 2 0` fails because Lean tries to infer $0 : \mathbb{N}$ first. The fix is to ascribe the base type: write `hasDerivAt_pow 2 (0 : ℝ)`, where $\mathbb{R}$ is the real-number type. Same pattern recurred several times in the closed-form derivative proof; documented in CLAUDE.md as a known type-ascription friction.

HasDerivAt.exp for $\exp \circ g$. The natural form of the chain rule for `exp` is `HasDerivAt.exp`, not `Real.hasDerivAt_exp.comp`. The latter doesn’t typecheck because `Real.hasDerivAt_exp` is a $\forall x, \text{HasDerivAt} \dots$ statement (a family of facts at each point), not a single `HasDerivAt` fact. The correct usage is `h.g.exp : HasDerivAt (exp ∘ g) (exp(g(0)) · g'(0)) 0` when `h.g : HasDerivAt g g'(0) 0`. Two minutes of search-and-replace once located.

Final ring closing a closed goal. At the end of `harmonic_linear_partition_eq`, the proof structure `congr 1; field_simp; ring` produced “No goals to be solved”. The `field_simp` had already closed the goal, so the trailing `ring` was redundant. Removed it. (This is the “`field_simp` sometimes closes the goal, sometimes leaves residue for `ring`” inconsistency already documented in the laplace CLAUDE.md.)

5 What was Mathlib, what was new

Mathlib provided.

- `integral_gaussian`: $\int_{\mathbb{R}} \exp(-b x^2) dx = \sqrt{\pi/b}$ for any $b \in \mathbb{R}$.
- Translation invariance of Lebesgue measure on $\mathbb{R}$.⁵
- `integral_const_mul` for pulling a constant out of an integral.
- `HasDerivAt.const_mul`, `HasDerivAt.div_const`, `HasDerivAt.exp`, `HasDerivAt.mul_const`, `hasDerivAt_pow`: standard `HasDerivAt`-algebra glue.
- `Real.sqrt_pos`, `Real.exp_zero`, `Real.exp_add`, `positivity`, `ring`, `field_simp`, `funext`.

Threepoint provided. The `GibbsRegularity` structure itself.

⁴It is closely related to the “integrand-lambda gymnastics” issue recorded in the anharmonic- κ_3 retrospective: when an integral’s integrand is a complicated expression, lambda-form rewriting plus `integral_congr` (or just substitution into the integral binding) is more robust than `conv_lhs => ext x`.

⁵The additive analogue (via `to_additive`) of `integral_mul_right_eq_self` in `MeasureTheory.Group.Integral`.

What was new.

- `harmonic_square_completion` (private): pointwise identity. Three lines.
- `exp_neg_t_harmonic_linear_eq` (private): pointwise factorisation of the Boltzmann factor. Six lines.
- `harmonic_linear_partition_eq` (public, headline helper): the closed-form partition for all h . Twenty-five lines.
- `harmonic_partition_eq` (public): $h = 0$ specialisation, $Z(0) = \sqrt{2\pi/(\lambda t)}$. Twenty lines.
- `harmonic_partition_pos` (public): the first `GibbsRegularity` field. Five lines.
- `integral_eq_zero_of_odd` (private): mirror of the helper from the harmonic- κ_3 tide. Eight lines. (Worth promoting to `lean-common` on this second use.)
- `harmonic_integral_neg_t_x_exp_neg_eq_zero` (public): the right-hand-side derivative integral, zero by parity. Six lines.
- `closed_form_hasDerivAt_zero` (private): derivative of the closed-form partition at $h = 0$ is zero. Twenty-five lines including the chain-rule plumbing.
- the headline structure instance⁶ (public): twenty lines.

The mathematical content is the closed form (an elementary calculation) and the parity-vanishing (already proven for the same weighted integrand in Tide 5, but on a different exponent parameterisation, so re-derived). Everything else is plumbing.

6 Lessons

- **Look for closed forms before the abstract machinery.** The pre-deliberation plan was to apply Mathlib’s dominated-convergence theorem; GPT’s pivot to the closed-form route saved an estimated 80–150 lines of plumbing and produced a stronger primitive. This generalises beyond the harmonic case: *whenever the partition or the moment integral is expressible in closed form, the closed form is the right primitive theorem to expose*, and the abstract regularity is a corollary. The dominated-convergence machinery is reserved for cases where no closed form is available.
- **The `conv => ext x` idiom does not work for integral integrands.** `conv_lhs => ext x; rw [...]` fails inside integral expressions because `ext` expects a function-typed target. Workaround: prove the integrand-equality at the function level (via `funext`-and-manipulation) and `rw` the resulting integral-level equality. Worth landing in `CLAUDE.md` as a recurring idiom alongside the “Pi.add vs single-lambda” note from earlier tides.
- **`HasDerivAt.exp`, not `Real.hasDerivAt.exp.comp`.** The chain rule for `exp ∘ g` is `h_g.exp`, with the value derivative $\exp(g(x_0)) \cdot g'(x_0)$. The pointwise `Real.hasDerivAt_exp` is a $\forall x$ statement and does not compose directly. Document.

⁶Full name: `Threepoint.harmonic_id.gibbsRegularity`.

- **The closed-form route’s stronger primitive is generic.** The route generalises to any potential of the form $L(x) = (\lambda/2)x^2 + Q(x)$ where Q admits a closed-form $\int e^{-tQ(x)}g(x) dx$ in terms of standard integrals — in particular to any potential whose Gibbs measure remains Gaussian after linear perturbation, or where the perturbation can be absorbed by substitution. For non-quadratic potentials (the anharmonic case from laplace, the bounded-prior cases, the sextic and pure-quartic from recent tides), no closed form for $Z(h)$ exists and the dominated-convergence route is the right one. Future tides that need `GibbsRegularity` for those cases will pay the dominated convergence cost.
- **Numerical sanity is sometimes not feasible, and that’s fine.** This tide’s target is structural (an instance of `GibbsRegularity`, a `Prop`-valued structure), not a numerical asymptotic. The skill’s discipline says to record “§Numerical check: not feasible: `!reason!`” rather than fake one; that is the right call here. The closed-form derivation is self-verifying via `integral_gaussian`, but it isn’t a numerical sanity check in the same sense as the anharmonic- κ_3 tide’s `scipy.integrate.quad` comparison.

7 Follow-ups

- *Cross-susceptibility derivative theorems for the harmonic case.* With the regularity instance now in hand, the FDT and cross-susceptibility derivatives specialise concretely. The cross-susceptibility derivative is particularly satisfying: by Tide 5, $\kappa_3(\varphi, x, B) = 0$ for the harmonic case, so

$$\partial_h \text{Cov}_h(\varphi, B)|_{h=0} = -t \cdot \kappa_3(\varphi, x, B) = 0$$

for any φ, B for which the relevant `GibbsObservable` hypotheses hold. A natural one-tide step would package this as a stand-alone theorem in `laplace`.

- *GibbsObservable instances for monomials.* The companion structure `GibbsObservable` (the φ -specific analytic content) is also still parameterised. Concrete instances for $\varphi(x) = x^k$ at the harmonic potential would follow the same closed-form route: the perturbed numerator $\int x^k e^{-t((\lambda/2)x^2 + h x)} dx$ has a closed form (it’s a derivative of $Z(h)$ in some other parameter, or a moment of the shifted Gaussian).
- *Promotion to lean-common.* `integral_eq_zero_of_odd` has now appeared in two files⁷ and is a self-contained generic statement. Worth promoting to `lean-common` as a small `Parity.lean` module on the next tide that needs it.
- *Stronger primitive: the moment-generating function.* The closed form $Z(h) = e^{th^2/(2\lambda)} \cdot \sqrt{2\pi/(\lambda t)}$ is, up to the $\sqrt{2\pi/(\lambda t)}$ normalisation, the *moment-generating function* of the Gaussian with variance $1/(\lambda t)$. Future tides that need higher harmonic moments can read them off from $Z(h)$ via differentiation; this is the natural generic primitive to expose.
- *Generalise to arbitrary at-most-linear A (Candidate C).* The dominated-convergence route would handle this; the closed-form route does not, since the partition is no longer Gaussian for nonlinear A . A future tide could implement the dominated-convergence route for the harmonic + general A case as a stepping-stone for non-Gaussian potentials.
- *Add the new gotchas to laplace/CLAUDE.md.* Three items: the `conv_lhs => ext x` non-applicability for integrals, the `HasDerivAt.exp-not-.comp` pattern, and the closed-form-first heuristic for `GibbsRegularity` instances.

⁷Once in `Threepoint/Harmonic.lean`, once here.

Acknowledgements

GPT-5.5 Pro contributed the deliberation-stage advisory at Step 2, including the strategic pivot from dominated convergence to the closed-form route, the corrected Young bound, and the verified suggestion to look for `integral_add_right_eq_self` and `integral_gaussian` as the load-bearing Mathlib calls. The full consult and the tide log itself are preserved in the patterning project's tide-log directory.⁸

The seabed for this tide spans threepoint's `GibbsRegularity` structure (Chris Elliott's earlier scaffold in `Threepoint/CrossSusceptibility.lean`), Mathlib's Gaussian integral and translation-invariance machinery, and the cross-repo dependency added by Tide 9 (`laplace [[require]]`-ing threepoint at commit 40398d8). The present tide is the first to actually *instantiate* threepoint's regularity bundle on a concrete potential, completing the loop that Tides 5 and 9 deliberately deferred.

⁸`sri/projects/patterning/tide-log/2026-05-06-tide-harmonic-gibbsregularity.md`
`gpt55_tide_harmonic_gibbsregularity.v1.md`.

and